3.1 Spot the Flaw

WORKSHEET

The question

STEP I 1991, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself When you are asked to "criticise each step", what should you check at each step? Make a list before you start.

STAGE 1 Is the derivative right?

Covers: Part (i)

Ask yourself Differentiate g yourself with the quotient rule, and factorise the numerator. Where is $g'(x) = 0$?

Pause and try Check $x = 1$ and $x = -1$ directly in your factorised $g'(x)$. Are they stationary points?

STAGE 2 Are the values right?

Covers: Part (i)

Ask yourself Work out $g(0)$, $g(1)$ and $g(-1)$ yourself. Do they agree with the argument?

STAGE 3 Does comparing values tell you which is a maximum?

Covers: Part (i)

Ask yourself Suppose a function has exactly two stationary points, with values $\frac{3}{4}$ and 1. Must the first be a minimum and the second a maximum?

Pause and try Decide the nature of the stationary point at $x = 0$ from the sign of $g'(x)$ on each side of 0.

STAGE 4 Do stationary values bound a function?

Covers: Part (i)

Pause and try Is the conclusion true anyway? What happens to $g(x)$ as $x \to \pm\infty$, and when is $g(x)$ negative? Look for an x with $g(x) < \frac{3}{4}$.

STAGE 5 The integral

Covers: Part (ii)

Ask yourself Differentiate $-3(1-x)^{-4}$. Do you get $(1-x)^{-3}$?

Ask yourself With the correct antiderivative, the answer still comes out as 0. So was the conclusion right after all?

Pause and try The fundamental theorem of calculus needs the integrand to be continuous on the whole interval. Is $(1-x)^{-3}$ continuous on $[-1,3]$? What

happens to the integral near the bad point?

What's really going on

When great mathematicians got it wrong

Errors of exactly this kind were made by the founders of analysis. In his *Cours d'analyse* of 1821, Augustin-Louis Cauchy "proved" that a convergent sum of continuous functions is always continuous. In 1826 Niels Henrik Abel pointed out a counterexample: the Fourier series

$$\sin x - \tfrac{1}{2}\sin 2x + \tfrac{1}{3}\sin 3x - \cdots$$

converges for every x, and every term is continuous, but the sum jumps at $x = \pm\pi$ (it equals $x/2$ for $-\pi < x < \pi$). Working out exactly what had gone wrong in Cauchy's argument took decades, and led to the concept of *uniform convergence*, which every analysis student now learns. Hidden assumptions, often about swapping the order of two limits, are the most common source of error in all of analysis.

Theorems with hypotheses

Two of the errors in this question come from ignoring the hypotheses of a theorem.

- The *extreme value theorem* says that a continuous function on a closed, bounded interval $[a, b]$ attains a maximum and a minimum. To find them, check the endpoints, the stationary points, and any points where the function is not differentiable: the minimum of $|x|$ on $[-1, 1]$ is at 0, which is neither. If f is differentiable everywhere inside the interval, an interior maximum or minimum must be a stationary point. On the whole real line there are no endpoints to check, and the function might not have a maximum at all; you must examine the behaviour as $x \to \pm\infty$ instead.
- The *fundamental theorem of calculus* needs the integrand to be continuous on $[a, b]$. If it is not, the integral may still exist (try $\int_0^1 x^{-1/2}\,\mathrm{d}x$) or may not (as here), and you must check using limits.

At university, every theorem you meet will come with a precise list of hypotheses. Part of the training is to check them, every time, before using the theorem.

A value for a divergent integral

There is a curious footnote to part (ii). The interval $[-1, 3]$ is symmetric about the bad point $x = 1$, and $(1 - x)^{-3}$ is "odd" about that point. If you cut out a symmetric gap $(1 - \varepsilon, 1 + \varepsilon)$ and let $\varepsilon \to 0$, the two infinite pieces cancel and you do get 0. This is called the *Cauchy principal value*, and it is genuinely useful in physics and complex analysis. But it is a *different definition*, chosen deliberately, and an argument that arrives at it by accident has not proved anything.

Try this next

1. **A substitution that lies.** Try STEP I 1987, Question 8 (step.maths.org/questions/87-s1-q8). A substitution seems to show that a positive integral equals a negative one. Find exactly where the argument breaks, then evaluate both integrals correctly.

2. **True or false?** Try STEP I 1998, Question 3 (step.maths.org/questions/98-s1-q3): four claims to prove or disprove. For each one, decide which you expect before you start writing.

3. **All horses are the same colour.** Find the flaw in this "proof" by induction: any one horse is the same colour as itself; given $n + 1$ horses, the first n are the same colour by the inductive hypothesis, and so are the last n; the two groups overlap, so all $n + 1$ are the same colour.

4. **An integral that does exist.** Show that $\int_{-1}^{1} |x|^{-1/2} \, \mathrm{d}x$ exists, although the integrand is undefined at 0, and find its value.

Further reading

- **E. A. Maxwell, *Fallacies in Mathematics*** (Cambridge University Press). A classic collection of plausible wrong arguments, organised by topic, written for sixth-formers.
- **Bernard Gelbaum & John Olmsted, *Counterexamples in Analysis*** (Dover). Hundreds of examples showing why the hypotheses of theorems matter. Dip in once you have started an analysis course.
- **Imre Lakatos, *Proofs and Refutations*** (Cambridge University Press). A brilliant dialogue about how mathematicians find, and fix, errors in proofs.

The full guided solution is in Section 3.1:
https://beyondstep.keithmcnulty.org/ch03/spot-the-flaw.html